# Investigation: Making the best can cooler




Aim:  To apply your knowledge of conduction, convection and radiation to make a can holder that keeps your cool drink cool for the longest.

Method:

In your group, use the various materials provided to construct a can holder that insulates (stops heat transfer) your drink from becoming warm too fast.

Judging criteria:

1. The can must be able to slip in and out of the holder with ease and not by being forced.
2. The cooler must be small enough to be gripped easily. If the cooler is so large it cannot be gripped, you will lose marks.
3. The cooler must be water proof, so if it is placed in water, the can cannot become wet. A fail in this criteria will mean your cooler fails overall.
4. The cooler must be a good insulator, which will be tested by placing a can in the cooler, with room temperature water. The can and cooler will then be placed in an esky with ice. The cooler which cools down the slowest means that it is the cooler that stops heat transfer the best.



Good luck.



1. Make your cooler out of the equipment provided. You have a set time limit, at the end of this time limit, you will have to stop, so make sure it is completed.
2. Test the cooler in a fair test by filling in the following work sheet.




**Investigation: Which cooler is best?**

Aim: To find out which cooler is the best insulator.

Independent variable: This is the variable you are deliberately changing. List the independent variable here;_____

Dependent variable: This is the variable you will be measuring. What is the dependent variable

_____

What will you use to measure it? Hint, you will need a stop watch and a

_____

Control variables: For your investigation to be a fair test, you need to keep all the other variables the same. List a minimum of 3 control variables

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Method:

1. Place ice in the esky, arrange all the coolers in the ice that are to be tested. Make sure no ice or cold water leaks into the coolers.
2. Place a thermometer in each cooler and add tap water to the can.
3. Record the temperature in your table.
4. Place the can into the cooler and start the stop watch.
5. After 10 minutes, record the temperature of the water in each can.
6. Place these results in your table.
7. Calculate the drop in temperature for each cooler.
8. Graph these results.
9. Answer the questions.

Results.



| Cooler (name of group) | Initial Temperature (°C) | Final Temperature (°C) | Drop in Temperature (°C) |
|---|---|---|---|
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Discussion:

1. Which cooler was the best?
2. Which cooler was the worse?
3. What design did the best cooler have that made it the best cooler? You should answer this question by talking about conduction, convection and radiation.

4. Were there any problems with the investigation, did anything go wrong or were any mistakes made that might have affected the results?

5. List any improvements you could make to this investigation if you did it again.